API NEWS SUMMARY

Project Progress Presentation

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March 28, 2026

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Introduction

This project focuses on developing an intelligent web application that extracts news articles from URLs and generates concise summaries. With the rapid growth of online content, users often struggle to read lengthy articles. This system aims to simplify information consumption by providing quick summaries, keyword extraction, and short URLs for easy sharing. The application supports multilingual news and converts them into English summaries for better accessibility.

The system integrates APIs for retrieving live news data and applies **Natural Language Processing (NLP)** techniques to generate summaries. This enables users to quickly grasp essential information without reading full articles. The application is designed to be user-friendly and efficient, providing summaries instantly based on user queries or selected categories.

Overall, the system aims to improve **information accessibility, efficiency, and user experience** by transforming lengthy news content into easily digestible summaries.

Literature Survey / Background Theory

Text summarization is an important area in Natural Language Processing that focuses on reducing large text into shorter versions while keeping the main meaning. With the increase in online news content, summarization systems have become useful to help users quickly understand information without reading full articles.

There are mainly two types of text summarization techniques: extractive and abstractive. Extractive summarization selects important sentences directly from the original text based on factors like word frequency and sentence importance. It is simple and fast but may not produce smooth or connected summaries. On the other hand, abstractive summarization generates new sentences by understanding the meaning of the text. It produces more natural summaries but is more complex and requires advanced models.

Many modern systems use deep learning models such as BERT, GPT, and T5. These models help in understanding context and improving the quality of summaries compared to traditional methods.

For news-based applications, APIs like NewsAPI are commonly used to fetch real-time articles. These APIs allow applications to collect news from different sources and categories, making it easier to build dynamic systems.

However, existing systems have some limitations. Many applications only display news without summarizing it, while others provide summaries but do not use real-time data. Some systems also face issues like loss of important information, repetition in summaries, and dependency on external APIs.

Based on this study, there is a need for a system that combines real-time news fetching with efficient summarization techniques. The proposed project aims to address this by integrating News API with summarization methods to provide quick and meaningful news summaries to users.

Problem Definition

The overwhelming volume of daily digital news creates information overload, making it difficult for users and businesses to identify relevant updates efficiently

1.There is a need for a system that can **automatically fetch real-time news and generate concise summaries**, enabling users to quickly understand key information.

Objectives:

- Fetch real-time news using APIs
- Generate concise summaries using NLP
- Reduce information overload
- Provide category-based news filtering
- Improve user reading efficiency

Methodology

The system follows these steps:

1. User logs into the application
2. User enters a news article URL
3. Article content is extracted using newspaper3k
4. If the article is in another language, it is translated into English
5. AI model (DistilBART) generates a summary
6. Slug and keywords are extracted from the URL/title
7. A short URL is generated for sharing
8. Results are displayed on the dashboard

Technologies used:

FastAPI (Backend)
HTML/CSS (Frontend)
Transformers (AI Model)
Newspaper3k (Article Extraction)

Work Done So Far

- Integrated News API for fetching articles
- Built backend structure
- Implemented initial summarization logic
- Developed basic UI
- Fixed issues related to short URL generation and summarization errors

Remaining Work

What Comes Next — Phase 2 Plan

- Improve accuracy of keyword extraction
- Optimize summarization speed
- Store short URLs in database for persistence
- Add category filters (Politics, Sports, Local)
- Enhance UI with animations
- Deploy the application online
- Add user authentication with database

Scope

The system can be extended for:

- Mobile applications
- Integration with real-time news APIs
- Personalized news recommendations
- Voice-based summarization
- Multi-language output summaries
- AI-based sentiment analysis

Conclusion

The News API Summarization system provides an effective solution to manage information overload by delivering concise and meaningful summaries of real-time news. By integrating APIs with NLP techniques, the system enhances user experience and accessibility to information. Future improvements can make it more intelligent and personalized.

References

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Thank You